NoisePrivacyMap<L1Distance<RBig>, PrivacyCurveDP> for ZExpFamily<1>

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This proof resides in **"contrib"** because it has not completed the vetting process.

This proves the privacy-map implementation for the discrete Laplace mechanism when the output measure is PrivacyCurveDP.

1 Hoare Triple

Precondition

MI is L1Distance<RBig> and MO is PrivacyCurveDP. The distribution scale is a non-negative rational.

Pseudocode

```python
class ZExpFamily1:
    def noise_privacy_map(self, _metric, _measure):
        scale = self.scale
        if scale < RBig.ZERO:  #
            raise "scale must be non-negative"

        def privacy_map(d_in):
            if d_in < RBig.ZERO:  #
                raise "sensitivity must be non-negative"
            if d_in == RBig.ZERO:  #
                return PrivacyCurve().with_approxDP([(0.0, 0.0)]).with_zCDP(0.0)
            if scale == RBig.ZERO:  #
                raise "no finite privacy guarantee"

            epsilon = f64.inf_cast(d_in / scale)
            sensitivity = f64.inf_cast(d_in)
            scale_f = f64.inf_cast(scale)
            if not epsilon.is_finite() or not sensitivity.is_finite() or scale_f <= 0:
                raise "privacy parameters are not finite"

            rho = zcdp_discrete_laplace(epsilon, sensitivity, scale_f)
            curve = PrivacyCurve().with_approxDP([(epsilon, 0.0)]).with_zCDP(rho)
            return curve.with_renyiDP(
                lambda alpha: rdp_discrete_laplace(alpha, sensitivity, scale_f)
            )

        return PrivacyMap.new_fallible(privacy_map)
```

Postcondition

Theorem 1.1. For every input distance accepted by the returned privacy map, the map returns a PrivacyCurve that is a valid upper privacy certificate for adding an independent discrete Laplace sample to each coordinate.

Proof. Lines 4 and 8 reject invalid parameters. With zero sensitivity, line 10 returns the identity certificate. With positive sensitivity and zero scale, the mechanism has no finite certificate and line 12 reports failure.

For positive rational sensitivity s and scale b, the discrete Laplace point-mass ratio calculation gives the pure-DP bound

$$\varepsilon = s/b.$$

The conservative casts in the pseudocode make the floating-point `epsilon` an upper bound on this value, so the $(\varepsilon, 0)$ point inserted into the curve is valid.

The implementation also computes the Harrison–Manurangsi zCDP bound using directed interval arithmetic. Therefore the returned ρ is no smaller than the exact zCDP bound. Finally, `rdp_discrete_laplace` evaluates the exact Rényi expression with directed intervals and falls back to the pure-DP bound if its numerics fail; either result is an upper bound at every order. Each representation is therefore a valid certificate, and querying or composing the resulting `PrivacyCurve` preserves conservativeness. $\square$